

Variability in the benefits of ocean acidification to photosynthetic rates of macroalgae without CO₂-concentrating mechanisms

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Abstract. Increasing concentrations of surface-seawater carbon dioxide (CO₂) (ocean acidification) could favour seaweed species that currently are limited for dissolved inorganic carbon (DIC). Among them, those that are unable to use CO₂-concentrating mechanisms (CCMs) to actively uptake bicarbonate (HCO₃[−]) across the plasmalemma are most likely to benefit. Here, we assess how the DIC uptake and photosynthetic rates of three rhodophytes without CCMs respond to four seawater CO₂ concentrations representing pre-industrial (280 μatm), present-day (400 μatm), representative concentration pathway (RCP) emissions scenario 8.5 2050 (650 μatm) and RCP 8.5 2100 (1000 μatm). We demonstrated that the photosynthetic rates of only one species increase between the preindustrial and end-of-century scenarios, but because of differing photosynthetic quotients (DIC taken up relative to O₂ evolved), all three increase their DIC uptake rates from pre-industrial or present-day scenarios to the end-of-century scenario. These variable, but generally beneficial, responses highlight that not all species without CCMs will respond to ocean acidification uniformly. This supports past assessments that, on average, this group will likely benefit from the impacts of ocean acidification. However, more concerted efforts are now required to assess whether similar benefits to photosynthetic rates and DIC uptake are also observed in chlorophytes and ochrophytes without CCMs.

Additional keywords: bicarbonate, carbon-concentrating mechanisms, carbon dioxide, carbon use, DIC use, non-CCM species, pCO₂.

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Introduction

Ocean acidification is the ongoing absorption of anthropogenically derived atmospheric CO₂ by surface seawater (Sabine *et al.* 2004). This results in increased concentrations of hydrogen ions (H⁺) and dissolved inorganic carbon (DIC). Additionally, there are changes in the speciation of DIC away from carbonate and in favour of carbon dioxide or carbonic acid (hereafter, CO₂) that is currently present at a low concentration, with slight increases in bicarbonate (HCO₃[−]) that currently forms the bulk of seawater DIC (Raven *et al.* 2005a). Seaweeds are important habitat-forming species in hard-bottom, photic zones throughout all oceans (Wernberg *et al.* 2019). Both kelp forest and coral-reef ecosystems are reliant on seaweed, and both systems support high biodiversity, and economic benefits such as fishing and tourism (Bennett *et al.* 2015; Comeau and Cornwall 2016; O'Mahony *et al.* 2017). Ocean acidification generally negatively affects the net calcification rates of calcareous seaweeds (Kroeker *et al.* 2013; Johnson *et al.* 2014; Cornwall *et al.* 2017a). However, the responses of non-calcareous

seaweed to ocean acidification are mixed, ranging from positive through to negative responses (Koch *et al.* 2013; van der Loos *et al.* 2019). Therefore, understanding generalities among the responses of non-calcareous seaweeds is more complex than is understanding those of calcareous seaweeds.

The response of non-calcareous (i.e. fleshy) seaweeds to ocean acidification is mediated by their use of DIC during photosynthesis (Cornwall *et al.* 2017b), and potentially altered by any impacts of changing intracellular pH caused by increased hydrogen ion/proton concentrations (hereafter, protons) in seawater (Raven 2013; van der Loos *et al.* 2019). The majority (~65%) of marine macroalgae possess CO₂-concentrating mechanisms (CCMs) that allow the energised uptake of HCO₃[−] (Stepien 2015). However, some species, mostly rhodophytes and chlorophytes, do not possess CCMs and are restricted to CO₂ uptake (Cornwall *et al.* 2015; Kübler and Dudgeon 2015; Stepien 2015). Hereafter, we refer to them as non-CCM species. It has been proposed that non-CCM species will generally benefit from the 180% increase in dissolved CO₂ projected

under representative concentration pathway (RCP) 8.5 by 2100 (Hepburn *et al.* 2011; Gattuso *et al.* 2015), which could result in elevated photosynthetic rates in any DIC-limited species. These predictions are manifested in the field, where the abundance of non-CCM species has been shown to increase at locations in Vulcano and Ischia, Italy, with naturally elevated CO₂ when compared to nearby reference locations with present-day concentrations of CO₂ (Cornwall *et al.* 2017b). However, although non-CCM species as a group increased in abundance, this was not the case for each individual non-CCM species.

The responses to ocean acidification and elevated CO₂ concentrations of only eight non-CCM species have been assessed in the laboratory (Table 1; Kübler *et al.* 1999; Cornwall *et al.* 2012; Gordillo *et al.* 2016; Kram *et al.* 2016; Britton *et al.* 2019; van der Loos *et al.* 2019). These species displayed a range of responses that indicate that the benefits of increasing CO₂ concentrations on the photosynthetic and growth rates of non-CCM species may vary. The inconsistency among these responses may be related to whether or not they are limited for DIC. When cultured under a saturating irradiance, species that are DIC limited would be expected to increase photosynthesis if CO₂ supply is increased, whereas species that are saturated for DIC would not (Hepburn *et al.* 2011). Additionally, all five of the tested six non-CCM species that did not increase their growth or photosynthetic rates, were exposed to only two different CO₂ concentrations. Therefore, differences among treatments could be difficult to detect because of either small effect sizes or because the photosynthetic rates of different species could saturate at different CO₂ concentrations. Assessing responses over multiple CO₂ concentrations at saturating irradiances could remedy these problems, and this was the goal of the present study.

Here, we assess the photosynthetic responses of three different shallow-living rhodophyte non-CCM species to four CO₂ concentrations, to test whether ocean acidification could benefit the photosynthetic rates of these species to differing degrees. We hypothesise that (1) elevated CO₂ concentrations would generally increase the photosynthetic rates–DIC uptake of all three species when assessed at saturating irradiance, but that (2) the magnitude of any benefit would differ between species.

Materials and methods

Fifty individuals each of three non-CCM rhodophyte macroalgae were collected from Tinderbox, Tasmania, Australia (43°03'31.8672"S, 147°20'0.3912"E) on 24 January 2015, using SCUBA at 3-m depth. These species were *Heminuera frondosa*, *Rhodomenia* sp. and *Plocamium angustum*. These species were each identified as a non-CCM species previously by pH-drift values lower than 9.0 and $\delta^{13}\text{C}$ values of lower than -30.0 , and were the three most common non-CCM species at the study site (Cornwall *et al.* 2015). Samples were placed into zip-lock bags underwater, then into black plastic bags before exiting the water. The black bags were placed into a cooler bin on land and transported back to the laboratory (3-min away), where they were placed into 0.5 μm -filtered (Whatman GF/C filters; GE Healthcare UK Ltd, Little Chalfont, UK) and UV-sterilised (Smart HO UV steriliser, 025050-2, 50-W lamp, Emperor Aquatics, Pottstown,

Table 1. Summary of previous responses of macroalgae without CO₂-concentrating mechanisms (non-CCMs) to elevated CO₂, approximating ocean acidification
Details include the species name, depth of collection, location of collection, irradiance and duration

Study	Species (and sample size)	Location of collection	Depth of collection (m)	Irradiance employed ($\mu\text{mol photons m}^{-2} \text{s}^{-1}$)	Duration (days)	Response (and carbonate chemistry)
Britton <i>et al.</i> (2019)	<i>Callophyllis lambertii</i> (6)	Tasmania	8–10	80	14	↑ net photosynthetic rates, no change in growth rates (pH 8.0 v. 7.64)
Britton <i>et al.</i> (2019)	<i>Plocamium dilatatum</i> (6)	Tasmania	8–10	80	14	No growth or photosynthetic-rate response (pH 8.0 v. 7.64)
Cornwall <i>et al.</i> (2012)	<i>Rhodophyllis gunnii</i> , now <i>R. membranacea</i> (5)	Southern New Zealand	0.5	230	15	No photosynthetic-rate response (pH 7.9 v. 7.5)
Cornwall <i>et al.</i> (2012)	<i>Schizoseris</i> sp. (5)	Southern New Zealand	0.5	230	15	No photosynthetic-rate response (pH 7.9 v. 7.5)
Kram <i>et al.</i> (2016)	<i>Plocamium cartilagineum</i> (4)	Southern California	Intertidal	Naturally variable, maximum ~9	21	No growth-rate response (pH 7.96 v. 7.75)
Kübler <i>et al.</i> (1999)	<i>Lomentaria articulata</i> (4)	Scotland	Unknown	40–50	21	↑ growth rates (350 v. 700 $\mu\text{mol mol}^{-1} \text{CO}_2$)
van der Loos <i>et al.</i> (2019)	<i>Craspedocarpus ramentaceus</i> (6)	Tasmania	6	25–30	7	No growth-rate response (pH 8.0 v. 7.70)

Table 2. Mean initial seawater carbonate chemistry

pH_T and dissolved inorganic carbon (DIC) were measured. Other components were calculated from the pH on the total scale (pH_T) and DIC with salinity of 35 and temperature 12°C. $n = 18$. Standard deviation or measurement error (if larger) in parentheses. A_T, total alkalinity; HCO₃⁻, bicarbonate; pCO₂, partial pressures of carbon dioxide

pCO ₂ treatment	pH _T initial	pH _T final	A _T (μmol kg ⁻¹)	DIC (μmol kg ⁻¹)	pCO ₂ (μatm)	CO ₂ (μmol kg ⁻¹)	HCO ₃ ⁻ (μmol kg ⁻¹)
280	8.15 (0.02)	8.16 (0.02)	2376 (5)	2087 (8)	276 (18)	11.4 (0.7)	1881 (19)
400	8.12 (0.02)	8.13 (0.02)	2410 (5)	2135 (9)	404 (28)	16.6 (1.2)	1972 (28)
650	7.93 (0.02)	7.95 (0.02)	2373 (5)	2199 (7)	658 (37)	27.1 (1.5)	2073 (37)
1000	7.58 (0.03)	7.66 (0.03)	2409 (5)	2264 (10)	1077 (77)	44.3 (3.2)	2154 (77)

PA, USA) seawater under constant aeration at 12°C and 50 μmol photons m⁻² s⁻¹ irradiance on a 12:12 h light–dark regime in a temperature-controlled room in 50-L aquaria. Samples were cleaned of epiphytes and left to acclimatise to the laboratory conditions for 48 h, before assessments of their photosynthetic rates began. Prior to these assessments, the maximum electron-transport rate (ETR_{max}) of a subset of individuals of each species was examined to determine the saturating irradiance. It determined that an irradiance of 100 μmol photons m⁻² s⁻¹ was sufficient to saturate photosynthetic rates under ambient pH (mean saturating irradiance = 104, only 2/10 replicates above 115, 4 replicates of *P. angustum*, 3 of the other 2 species). Salinity was measured with a YSI conductivity probe and was 35 ± <0.1 (mean ± s.e.).

To assess the photosynthetic rates of all three species, we used a randomised treatment-level design with six individuals. To examine the effects of CO₂, we used methods that chemically simulated the effects of ocean acidification (Hurd *et al.* 2009; Riebesell *et al.* 2010), while retaining appropriate replication of all treatments (Cornwall and Hurd 2016). We created 10-L batches of seawater that were modified to simulate a pre-industrial treatment by bubbling of N₂ gas to increase pH, and then bubbling with O₂ gas to increase O₂ concentrations back to ambient. We then reduced the seawater pH and increased CO₂ for the other three treatments by using equal additions of 0.1 mol HCl and NaHCO₃ in one 1-L container for each replicate, a processes chemically identical to ocean acidification (Riebesell *et al.* 2010; Cornwall *et al.* 2013), i.e. 72 independent batches in total. The four treatments used represented environmentally relevant partial pressures of carbon dioxide (pCO₂) values of 280 (pre-industrial), 380 (present-day), 650 (mid-century RCP 8.5) and 1000 μatm (end-of-century RCP 8.5). See Table 2 for relevant seawater carbonate chemistry.

A subsample of 1-g wet weight of each of six replicate samples of the three species was placed into 650-mL Perspex chambers described in Britton *et al.* (2016). The initial and final pH on the total scale (pH_T; hereafter ‘pH’) was measured using a Tris and amp buffers (Dickson *et al.* 2007; see Table 1), and initial and final O₂ concentrations were measured with an Orion RDO probe (ORI087100MDW, ThermoFisher Scientific, Waltham, MA, USA). The RDO probe was calibrated using 100 and 0% standards, i.e. seawater was aerated with 100% O₂ or with N₂ respectively, for 15 min. Samples were taken for initial and final DIC from each replicate. Dissolved inorganic carbon was measured using an

Apollo SciTech DIC analyser model AS-C3 (Apollo SciTech, Newark, DE, USA) with an inbuilt LI-COR LI-7000 CO₂/H₂O analyser (LI-COR, Lincoln, NB, USA). Dissolved inorganic carbon measurements were calibrated against a certified reference material provided by Andrew Dickson (Scripps Institute for Oceanography, San Diego, CA, USA). Specimens were left under irradiance of 100 μmol photons m⁻² s⁻¹ for 2 h. Trials without seaweed indicated minimal changes in both oxygen concentration and pH that were within the measurement error. At the end of incubations, dry weights were measured after 24 h in a drying oven at 60°C. Rates of DIC and O₂ change per hour were then standardised to dry weights of the samples. Total alkalinity (A_T) and pCO₂ were calculated using the constants of Mehrbach *et al.* (1973) and the refit by Dickson and Millero (1987). These analyses were performed using the CO₂ calculator app developed by Professor Keith Hunter (see Orr *et al.* 2015). The photosynthetic quotient (PQ) was also calculated as oxygen release divided by DIC uptake (Axelsson 1988).

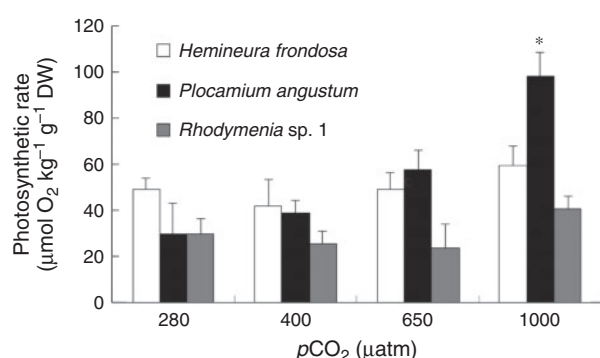
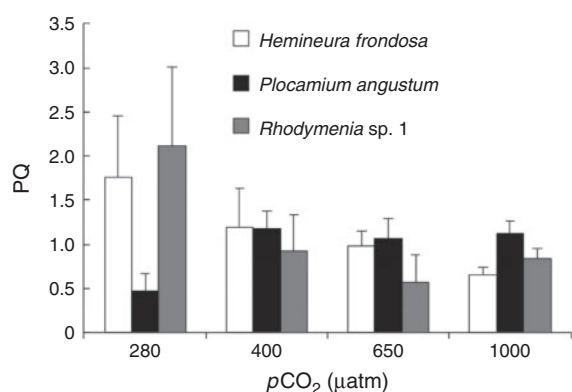
We analysed the effect of treatment levels on photosynthetic rates, DIC uptake and PQ within each species by using a general linear model with discrete factors, followed by Tukey honest significant difference *post hoc* tests, to test the specific hypothesis that species would display elevated rates at different CO₂ concentrations. This model choice was then compared with a model using CO₂ as a continuous factor. No changes in the detection of significant effects of CO₂ were found between the two different model choices. All analyses were performed in R (ver. 3.5.1, R Foundation for Statistical Computing, Vienna, Austria).

Results

Photosynthetic rates of *H. frondosa* and *Rhododymenia* sp. species were not significantly altered by the different treatments, whereas those of *P. angustum* were higher than all others in the 1000-μatm treatment (Table 3, Fig. 1). However, even though the PQ was not statistically different between the different pCO₂ levels for any species (Table 2, Fig. 2), there was considerable variability between individuals and pCO₂ levels. Dissolved inorganic carbon uptake in all three species was significantly higher in the 1000-μatm treatment than it was in the 280 and 400 treatments (Table 3, Fig. 3). This equated to a 180% increase in DIC uptake between the control treatment (400) and the end-of-century (1000) treatment for *H. frondosa*, 250% increase for *Rhododymenia* sp. and 280% increase for *P. angustum*.

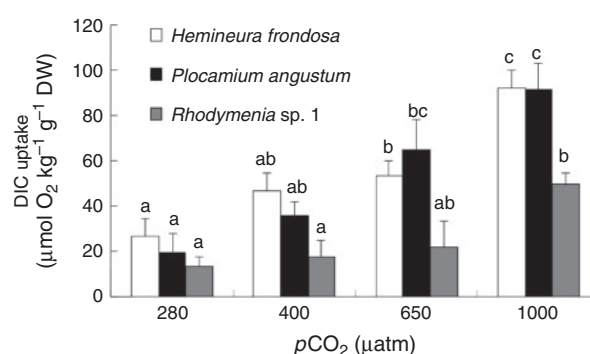
Table 3. Analysis of variance table for photosynthetic rate, dissolved inorganic carbon (DIC) uptake, and photosynthetic quotient (PQ)*F*-value and *P*-value are presented. *P*-values <0.05 are in bold

Species	Rate	<i>F</i> -value	<i>P</i> -value
<i>Hemineura frondosa</i>	Photosynthetic rate	0.789	0.514
	DIC uptake	13.280	<0.001
	PQ	1.174	0.345
<i>Plocamium angustum</i>	Photosynthetic rate	9.407	<0.001
	DIC uptake	9.769	<0.001
	PQ	2.820	0.065
<i>Rhodymenia</i> sp. 1	Photosynthetic rate	1.082	0.379
	DIC uptake	4.895	0.010
	PQ	1.717	0.196

**Fig. 1.** Photosynthetic rates of three species without CO₂-concentrating mechanisms (non-CCMs) under four partial pressures of carbon dioxide (*p*CO₂) levels. Asterisk indicates rates significantly (*P* < 0.05) different from others. Bars represent means ± standard errors (*n* = 6). DW, dry weight.**Fig. 2.** Photosynthetic quotient (PQ), the rate of oxygen release divided by dissolved inorganic carbon (DIC) uptake, of three species without CO₂-concentrating mechanisms (non-CCMs) under four partial pressures of carbon dioxide (*p*CO₂) levels. Means with standard errors in parentheses. *n* = 6 per treatment level and species combination. Bars represent means ± standard errors.

Discussion

Here, we have highlighted that non-CCM seaweed species will tend towards an increasing DIC uptake under ocean acidification predicted by the end of the century under RCP 8.5, but the extent

**Fig. 3.** Dissolved inorganic carbon (DIC) uptake rates of three species without CO₂-concentrating mechanisms (non-CCMs) under four partial pressures of carbon dioxide (*p*CO₂) levels. Within each species, bars not sharing letters are significantly (*P* < 0.05) different from each other. Bars represent means ± standard errors (*n* = 6). DW, dry weight.

of measurable increases in photosynthetic rates are variable and species-specific. Photosynthetic rates of all three species were highest under 1000 μatm *p*CO₂, but the strength of these effects were overwhelmed by the stronger role of individual variability in oxygen-evolution rates. These results support past hypotheses regarding the benefits that this seaweed group could receive from ocean acidification (Hepburn *et al.* 2011; Cornwall *et al.* 2012, 2015, 2017b), while reinforcing the importance of -species variability within this group (Cornwall *et al.* 2012; van der Loos *et al.* 2019). However, because of the moderate replication here and in all past studies such as this (Table 1), such variability could also be influenced by the power to detect these effects.

We know little about how non-CCM species will respond to ocean acidification, and even less is known about their DIC physiology, despite their high species number and abundance in some locations (e.g. Cornwall *et al.* 2015). Van der Loos *et al.* (2019) advocated that species without CCMs might not necessarily be DIC limited. Although DIC uptake in these species does not occur by active HCO₃⁻, there are other mechanisms that they could employ to increase DIC uptake. The activity of external carbonic anhydrase is higher among *Caulerpa* spp. that do not have CCMs, for example, than it is with those that do (Kevekordes *et al.* 2006). Therefore, non-CCM species would be expected to have varying DIC affinities

on the basis of their activity of external carbonic anhydrase, or potentially the numbers of aquaporins or content of Rhesus proteins, which could increase movement of CO₂ across the plasmalemma (Raven *et al.* 2005b; Kustu and Inwood 2006). Differential 'leakiness' of CO₂ among species could also influence responses to CO₂, where leakage in some species could decrease with an increasing CO₂ concentration (Raven *et al.* 2012, 2014; Raven and Beardall 2014; Cornwall *et al.* 2017b). Furthermore, the interaction between seaweed morphology and water velocity might affect their responses to the elevated CO₂ concentration. Species with morphologies that are prone to thicker diffusion-boundary layers, which can cause a mass transfer limitation of DIC uptake, could respond differently to the same CO₂ concentrations than nearby species with thinner diffusion-boundary layers. These are all possible explanations for differential responses to ocean acidification related to DIC uptake.

It is also possible that responses to ocean acidification could be further modified by pH or proton concentration itself (Raven 2013; van der Loos *et al.* 2019). External seawater pH will alter pH within the seaweed; for example, external seawater strongly influences pH at the site of calcification within coralline algae (Cornwall *et al.* 2017a; Comeau *et al.* 2018). Although pH within the cytoplasm is much lower (~7.5, Ritchie 1985), ocean acidification would likely still cause reductions by increasing proton-concentration gradients between internal and external seawater (Raven 2013). This will have unknown consequences for enzyme activity and other cellular functions. For example, narrow pH limits are required for the optimal activity of key enzymes such as phosphoenolpyruvate carboxylase in terrestrial plants (Smith and Raven 1979).

Predicting the future responses of non-CCM seaweeds to ocean acidification is complex; however, the increases in DIC uptake observed here in three species corroborated the hypothesis that, as a group, non-CCM seaweeds may benefit. This has been observed by a general increasing abundance of both freshwater and seawater non-CCM macroalgae as a group in response to elevated CO₂ concentrations at naturally elevated CO₂ sites (Maberly *et al.* 2015; Cornwall *et al.* 2017b), and by increased growth (Kübler *et al.* 1999) and DIC uptake (Cornwall *et al.* 2012) in the laboratory. At the same time, our observations also support the fact that variability among the responses of non-CCM species to ocean acidification exists, and benefits might not always be expected to occur in every one of these species (e.g. both Kram *et al.* 2016; van der Loos *et al.* 2019). Each of these prior laboratory assessments focused on rhodophytes; the question now is how chlorophyte and ochrophytes without CCMs could respond to simulated ocean acidification. Although they are rarer than are rhodophyte non-CCM species, this is particularly important considering the large abundance of *Caulerpa* spp. without CCMs in some ecosystems in Australia (Cornwall *et al.* 2015) and New Zealand (C. E. Cornwall, pers. obs.), and their larger size than that of rhodophytes, and, hence, a larger contribution of biomass to the ecosystems they inhabit.

Conflicts of interest

The authors declare that they have no conflicts of interest.

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